

**Module 3: Critical Thinking**  
**Option 2: Predicting Fuel Efficiency Using Tensor Flow**

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August 9, 2026

# **Predicting Automobile Fuel Efficiency Using TensorFlow Regression**

## **Introduction**

Deep learning has become a powerful approach for solving regression problems where the objective is to predict a continuous numerical value. Unlike classification problems that assign data to categories, regression models estimate values such as housing prices, product sales, or vehicle fuel efficiency. In this project, TensorFlow and Keras were used to develop a regression model capable of predicting the miles per gallon (MPG) of late-1970s and early-1980s automobiles using the Auto MPG dataset. Vehicle characteristics, including the number of cylinders, engine displacement, horsepower, weight, acceleration, model year, and country of origin, served as the model's input features. The objective was to examine the complete deep learning workflow, including data preparation, model construction, training, evaluation, and prediction using a neural network.

## **Data Preparation and Exploration**

The Auto MPG dataset was downloaded directly from the UCI Machine Learning Repository using TensorFlow's `keras.utils.get_file()` function and imported into Python using the Pandas library. Initial exploration of the dataset confirmed the presence of vehicle characteristics alongside their corresponding fuel efficiency values. Figure 1 presents the final rows of the imported dataset.

During data inspection, six missing horsepower values were identified. Because these represented only a small percentage of the total observations, the incomplete records were removed using Pandas' `dropna()` function. This reduced the dataset from 398 to 392 observations while preserving sufficient data for model training.

The cleaned dataset was divided into training and testing datasets using an 80/20 split. Exploratory data analysis was then performed using a pairplot to visualize relationships among several variables. As shown in Figure 2, vehicle weight and engine displacement appear to have inverse relationships with fuel efficiency, suggesting these variables are likely to be important predictors.

Descriptive statistics were calculated for each input feature (Figure 3). These statistics were later used to normalize the dataset by subtracting each feature's mean and dividing by its standard deviation. Normalization ensures that variables measured on vastly different scales, such as vehicle weight and cylinder count, contribute equally during model training.

## **Neural Network Development and Training**

A TensorFlow Sequential regression model was constructed using two hidden dense layers containing 64 neurons each with ReLU activation functions. The output layer consisted of a single neuron with a linear activation function for predicting MPG. Figure 4 summarizes the architecture of the completed neural network.

The model was compiled using the RMSprop optimizer with Mean Squared Error (MSE) as the loss function. During training, both Mean Absolute Error (MAE) and Mean Squared Error were monitored as evaluation metrics. Before training began, the model was tested using a small batch of normalized input data to verify that predictions could be generated successfully.

The model was then trained for 1,000 epochs using 80% of the training data, while 20% of the training data was reserved for validation. TensorFlow stored the training history throughout the learning process, allowing changes in loss and error metrics to be monitored across all epochs. Figure 5 presents the final training history after completion of the training process.

Figures 6 and 7 illustrate the learning curves for Mean Absolute Error and Mean Squared Error, respectively. Both metrics decreased substantially during training, indicating that the neural network successfully learned relationships between vehicle characteristics and fuel efficiency. Although the validation error remained higher than the training error, the model demonstrated reasonable generalization without severe divergence between the two curves.

## **Model Evaluation**

Mean Absolute Error (MAE) and Mean Squared Error (MSE) provide complementary perspectives for evaluating regression performance. MAE reports the average prediction error in the same units as the target variable (MPG), making it intuitive to interpret. MSE squares prediction errors before averaging them, causing larger prediction errors to receive greater penalties. Consequently, MSE is particularly useful when large prediction mistakes should be minimized.

Because this project involved predicting a continuous value rather than classifying observations, a regression neural network was an appropriate modeling choice. While simpler approaches such as linear regression could also predict MPG, the neural network provides greater flexibility for learning nonlinear relationships that may exist among multiple vehicle characteristics.

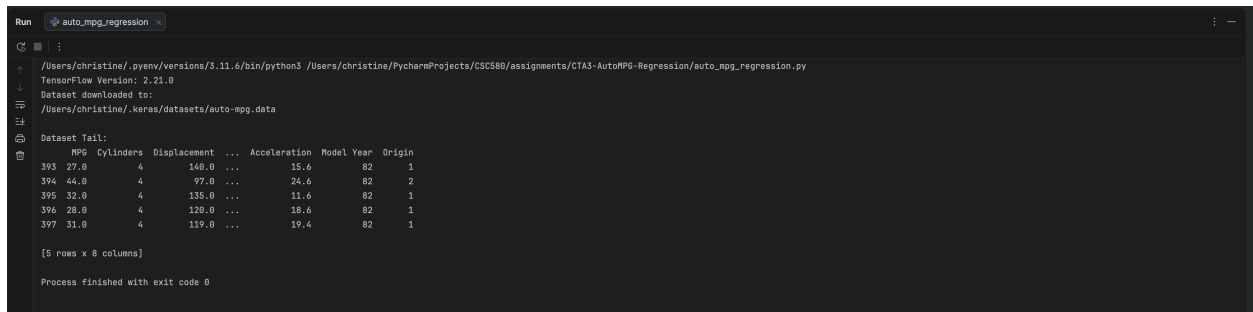
## **Conclusion**

This assignment demonstrated the complete process of developing a regression model using TensorFlow, from preparing raw data through training and evaluation. Beyond predicting automobile fuel efficiency, the techniques explored in this project provide a foundation for more complex regression problems involving healthcare and longitudinal patient monitoring.

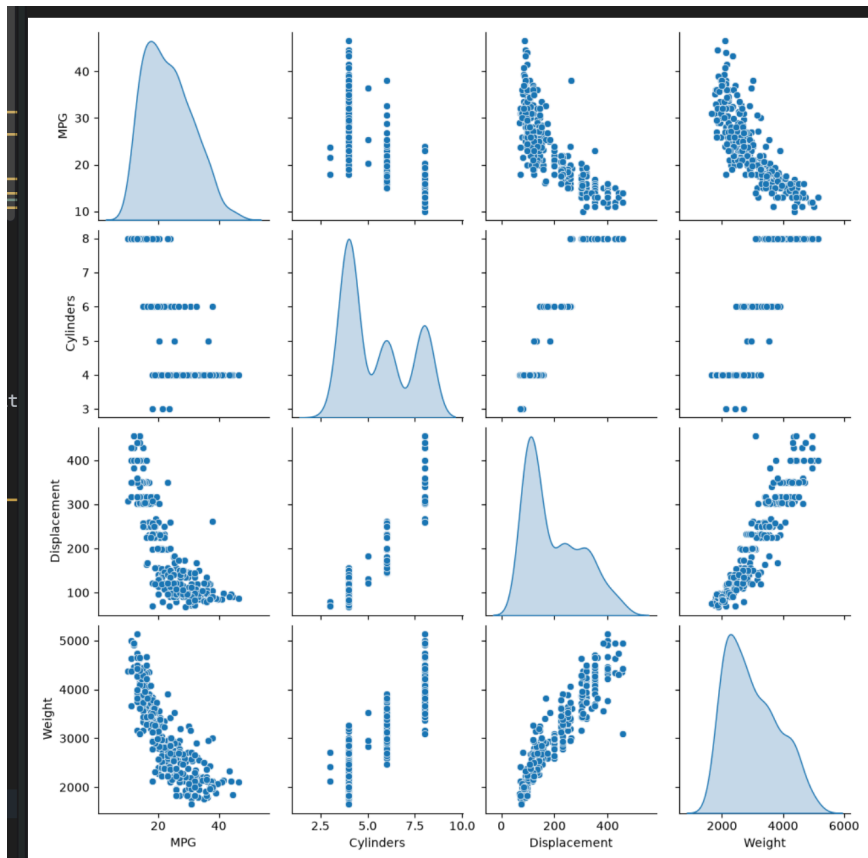
Within my nonprofit, AuVentures, I envision applying similar regression and deep learning techniques to estimate disease severity and predict autoimmune flare risk using longitudinal patient information. Although real-world clinical prediction requires substantially more sophisticated approaches, including temporal modeling methods such as Dynamic Time Warping (DTW), recurrent neural networks, and transformer-based architectures, this project reinforces the fundamental workflow shared by many machine learning systems: prepare high-quality data, normalize meaningful features, train predictive models, evaluate performance, and continually improve prediction accuracy through iterative refinement.

## Figures

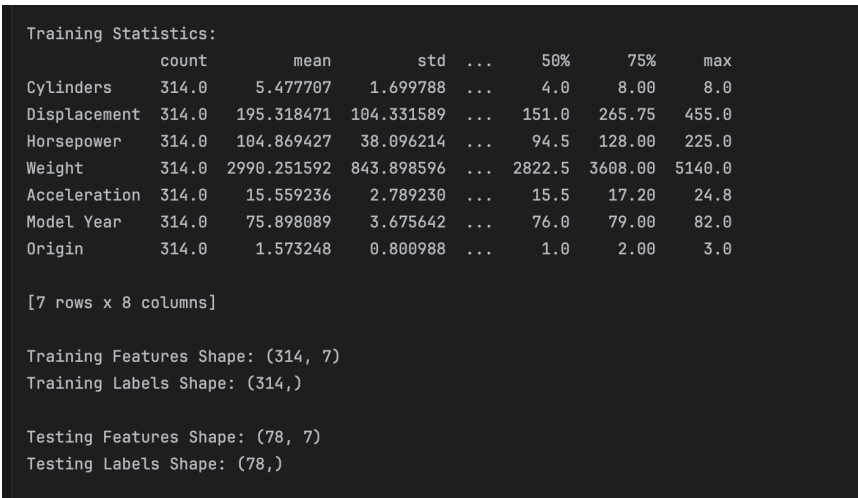
**Figure 1.** Last five records of the Auto MPG dataset after loading into a Pandas DataFrame. The dataset contains vehicle characteristics including cylinders, displacement, horsepower, weight, acceleration, model year, and origin. Miles per gallon (MPG) serves as the target variable for the regression model.



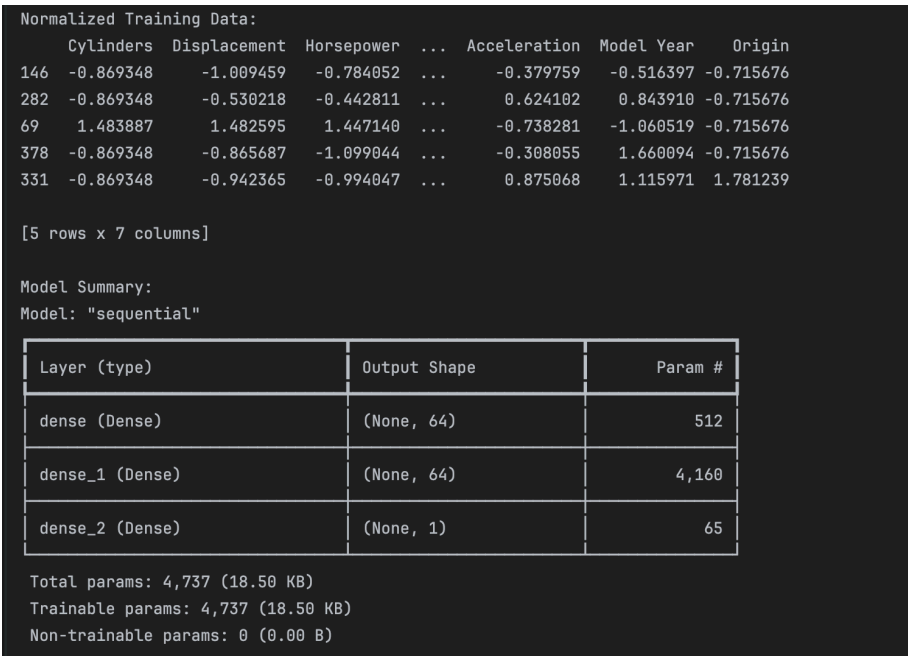
**Figure 2.** Pairplot showing relationships between MPG, cylinders, displacement, and vehicle weight in the training dataset.



**Figure 3.** Descriptive statistics for the training dataset after data cleaning. The table summarizes the distribution of each input feature, including the mean, standard deviation, and range used later for data normalization.



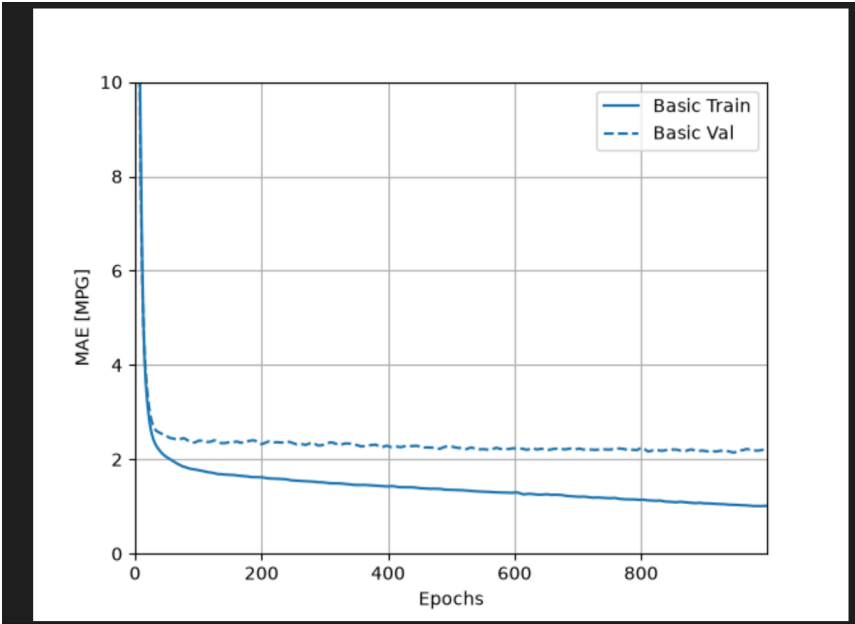
**Figure 4.** Summary of the TensorFlow regression model architecture. The model consists of an input layer with seven normalized features, two fully connected hidden layers containing 64 neurons each with ReLU activation functions, and a single linear output neuron used to predict miles per gallon (MPG).



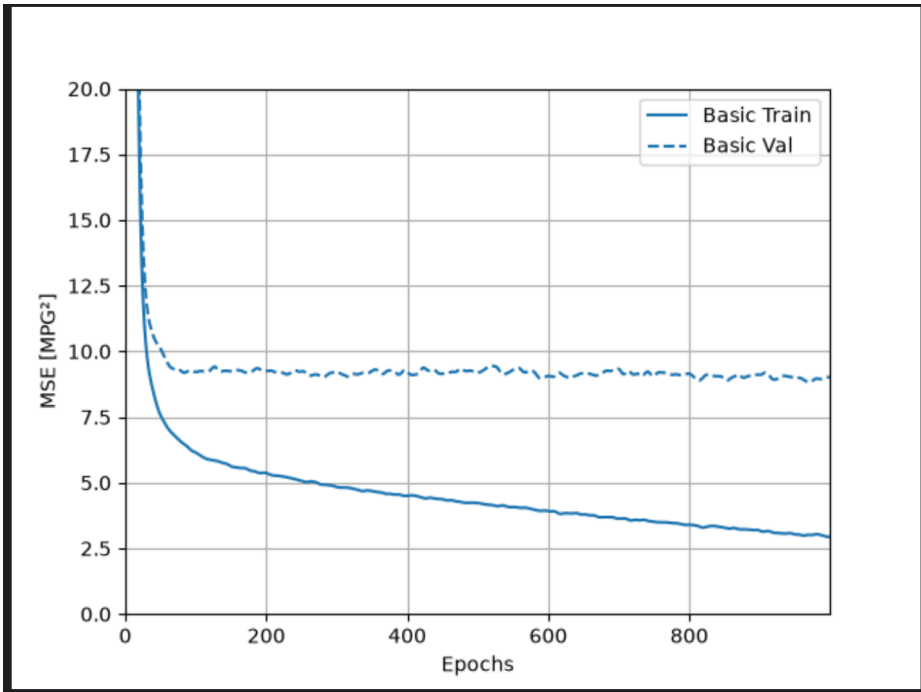
**Figure 5.** Final training history showing the last five epochs of model training. The table reports the training and validation loss, Mean Absolute Error (MAE), and Mean Squared Error (MSE) after 1,000 training epochs.

| .....             |          |          |          |          |          |          |       |
|-------------------|----------|----------|----------|----------|----------|----------|-------|
| Training History: |          |          |          |          |          |          |       |
|                   | loss     | mae      | mse      | val_loss | val_mae  | val_mse  | epoch |
| 995               | 2.356333 | 0.973337 | 2.356333 | 8.719920 | 2.298741 | 8.719920 | 995   |
| 996               | 2.303035 | 0.962635 | 2.303035 | 9.512102 | 2.415533 | 9.512102 | 996   |
| 997               | 2.368354 | 1.000227 | 2.368354 | 8.729125 | 2.287815 | 8.729125 | 997   |
| 998               | 2.549819 | 1.009101 | 2.549819 | 9.115946 | 2.374738 | 9.115946 | 998   |
| 999               | 2.601398 | 1.016107 | 2.601398 | 8.536740 | 2.283313 | 8.536740 | 999   |

**Figure 6.** Mean Absolute Error (MAE) during training and validation over 1,000 epochs. The decreasing error indicates that the neural network improved its prediction accuracy as training progressed.



**Figure 7.** Mean Squared Error (MSE) during training and validation over 1,000 epochs. The decreasing trend demonstrates improved model performance while illustrating differences between training and validation error throughout the learning process.



## References

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